MILITARY SPECIFICATION

FIBER, ACRYLIC, FIBRILLATABLE

1. SCOPE

1.1 This specification covers one type of fibrillatable acrylic fiber.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

TAPPI METHOD Technical Association of the Pulp and Paper Industry

T1002 - Drainage Time of Pulp for Insulating Board

(Application for copies should be addressed to the Secretary, TAPPI, 360 Lexington Avenue, New York 17, N.Y.)

FSC: 1320

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3. REQUIREMENTS

3.1 Materials.- The material shall be a fibrillatable gelled acrylic fiber, spun by the wet process, consisting of not less than 85% polyacrylonitrile with methyl methacrylate copolymer. The material shall contain no delusterants and shall comply with the requirements of all applicable drawings and specifications.

3.2 Denier.- The denier of the acrylic fiber shall be 5.4 ± 0.5 when tested in accordance with 4.3.3.1.

3.3 Fiber length.- The fibers shall have an average cut length of 3/8 inch, with the following distribution about the mean, when tested in accordance with 4.3.3.2.

CELL COUNT	%
1/16 to 3/16	14.0 max
4/16 to 9/16	79.5 min
10/16 to 12/16	6.0 max
greater than 12/16	0.5 max

3.4 Water content.- The acrylic fiber shall contain a minimum of 40 percent water when tested in accordance with 4.3.3.3.

3.5 Workmanship.- The acrylic fibers shall be uniform in quality and shall be free of impurities and other defects that could adversely affect their use.

3.6 Fibrillation.- The acrylic fiber shall be fibrillatable to a drainage rate of 12.5 ± 2.5 seconds when tested in accordance with 4.3.3.4. Microphotographs of the fiber before and after fibrillation shall be used to verify that the fiber is a fibrillating grade.

3.7 First Article Sample.- This specification makes provisions for submittal of a first article sample. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

4.1.1 Submission of product.- At the time the completed lot of material deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with an identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Quantity of material in the lot.
e. Date submitted.
f. Number of filaments present in tow band.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

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4.2.1 Submission.- The contractor shall submit a first article sample consisting of a minimum of 40 pounds (oven-dry basis) of fibrillatable acrylic fiber and a tow sample (10 meters) which have been produced by the same production processes which the contractor will use in fulfilling the contract, to a Government facility designated by the contracting officer. A similar sample shall also be submitted whenever a change occurs in the manufacturing process, material used, specification or source of supply, such as to significantly affect product uniformity as determined by the Government. A change in Merge No. resulting from above changes shall necessitate resubmittal of first article samples (see 6.4).

4.2.2 Inspections to be performed.- The material shall be inspected for all the requirements of this specification.

4.2.3 Rejection.- If the material fails to comply with the applicable requirements, the initial production quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new first article sample, or portion thereof, as specified by the contracting officer. The contractor shall continue to submit new or supplemental quantities as necessary until such time as a quantity successfully passes the inspection specified or until otherwise directed by the procuring activity.

4.3 Inspection provisions

4.3.1 Lot formation.- The term "inspection lot" as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Each lot shall consist of drums of fibrillatable acrylic fiber having identical Merge Numbers that have been subjected to the same processing operations and conditions, by one manufacturer, and in accordance with the same specification and specification revision.

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4.3.2 Examination.- The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the fiber designated in the applicable paragraph headings. A sufficient quantity of the finished fiber shall be drawn from each 20th drum of product, prior to sealing the plastic liner, to provide samples for lot examination and testing (see 6.3).

4.3.2.1 Fibers

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:			
101.	Luster	Visual	01001
Minor:			
201.	Workmanship (see 3.5)	Visual	01002

4.3.2.2 Fibers after packaging (see 5.1).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:			
101.	Liner torn	Visual	02001
102.	Liner improperly sealed	Visual	02002
Minor:			
201.	Marking incorrect, misleading, or unidentifiable	Visual	02003
202.	Workmanship (see 3.5)	Visual	02004

4.3.3 Testing

4.3.3.1 Denier (see 3.2), Major defect, Code No. 03001.- The denier of the acrylic fiber shall be determined in accordance with 4.4.1. Duplicate samples of acrylic tow shall be tested for each lot. If the average of the two samples fails to meet the applicable requirement, the lot shall be rejected.

4.3.3.2 Fiber length (see 3.3), Major defect, Code No. 04001.- The fiber length shall be determined for each lot in duplicate on randomly selected fiber chips as specified in 4.4.2. Failure to comply with the applicable requirement shall be cause for rejection of the lot.

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4.3.3.3 Water Content (see 3.4), Major defect, Code No. 05001.- Each 20th drum in the lot shall be randomly sampled in duplicate after cutting and prior to packing. Each sample shall be 100 grams minimum and shall be placed into wide-mouth glass jars (with screw cap). Close the jar immediately to prevent loss of water. Testing shall be as specified in 4.4.3. If either sample fails the applicable requirement, all of the acrylic fiber produced since the last previous test for water content shall be rejected. Each succeeding drum of fiber produced shall be sampled in duplicate and tested until five successive drums have met the requirement.

4.3.3.4 Fibrillation (see 3.6), Major defect, Code No. 06001.- Two samples from each lot of the cut fiber of 3200 grams (oven dry basis) each shall be randomly sampled and tested for fibrillation as specified in 4.4.4. Failure of either sample to fibrillate to the specified drainage rate or for the microphotographs to verify fibrillation shall be cause to reject the lot.

4.4 Test Methods and Procedures.

4.4.1 Denier.- Each sample of approximately two meters in length shall be accurately measured to the nearest 0.001 meter. Dry the measured samples for one hour minimum at 160 ± 2°F. Further condition the samples for four hours minimum at 23 ± 2°C and 50 ± 2% Relative humidity. Weigh each sample to the nearest 0.0001 gram. Average filament denier shall be calculated by the following equation:

$$\text{Average filament denier} = \frac{9000 \frac{A}{B}}{C}$$

Where:

A = Total weight of dried filaments in grams.

B = Total length of filaments before drying in meters.

C = Number of filaments.

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4.4.2 Fiber Length.- The sorting and weighing of the fibers must be accomplished as quickly as possible to assure that the fibers do not become dry. Transfer a quantity of the wet fiber to a wire screen fabricated from 1/2 inch screening. Shake a quantity of the fiber chips onto a black cloth which has been previously marked in a 2 inch grid patern. (A chip is defined as a small bundle of fibers in which the ends of the fibers are clearly defined. If the fibers in a chip have been separated such that both ends of the chip are not clearly defined, discard that chip and select another). Using tweezers, pick up all measurable chips from within certain squares in the grid. The squares selected are pre-determined on a random basis. Using a rule with 1/16 inch graduations, measure each chip to the nearest 1/16 inch. Group the chips in the four cells of the stated requirement (see 3.3). Repeat the above procedures until a total sample weight of 1 $\pm$ 0.1 gram (total of the four cells) is obtained. Weigh each cell to the nearest 0.001 gram. Calculate and report the weight percent of the four cells.

4.4.3 Water content.- Transfer the wet fiber sample from the closed jar to the weighing pan of a suitably calibrated moisture determination balance. Record the weight of the wet sample to the nearest 0.1 gram. Dry the fiber sample to constant weight using the heat source provided with the balance. Record the dry weight of the sample to the nearest 0.1 gram. Calculate the moisture content of the sample to the nearest 0.5 percent.

4.4.4 Fibrillation.- Gradually charge 90 liters of distilled or deionized water and 2500 $\pm$ 5 grams (Oven-dry basis) of the fiber to a clean, 5-pound Valley Niagara type heater. Addition to the beater will require approximately one hour to maintain circulation of the mixture. After 1500 grams of the fiber have been added, place an 8-pound weight on the bedplate lever arm of the beater. Add a 3-pound weight after addition of 2500 grams of fiber. Gradually add an additional 700 $\pm$ 5 grams (oven-dry basis) of the fiber to the beater. If required to assure circulation, an additional ten liters of distilled or deionized water may be added. This addition of fiber and water may require as long as 30 minutes to charge the entire 700 grams of fiber and still maintain stock circulation in the beater. When beater circulation is stabilized, add an additional 5-pound weight to the bedplate lever arm (total weight is now 16 pounds). Record the time as the start of the beating operation. Maintain the stock temperature, throughout the beating operation, at less than 100°F. Record the stock temperature and the load on the beater motor at 1/2-hour intervals. Continue beating until the drainage rate specified is attained, as determined in accordance with TAPPI Method T1002. Record total beating time. Microphotographs of the fiber before and after fibrilation should appear as shown in Figures 1A and 1B, respectively.

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5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level C.- The product shall be packed to afford protection against damage and contamination during direct shipment from the supply source to the first receiving activity for immediate use.

5.2 Marking.- Marking shall be in accordance with MIL-STD-129. In addition, each drum of product shall be marked with the dry weight of the contents.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the title, number, revision and date of this specification and Merge Number.

6.2 Inspection code numbers.- The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Denier.- Samples for fiber denier shall be selected from the acrylic tow prior to cutting.

6.4 New product approval.- One type of fibrillatable acrylic fiber which has been found satisfactory for the intended purpose is T-98 Cyanamide Acrylic Fiber manufactured by American Cyanamid Co., Fibers Division, Wayne, N.J. Use of similar materials from other sources must be approved by Picatinny Arsenal, ATTN: SMUPA-ND2, Dover, N.J. 07801

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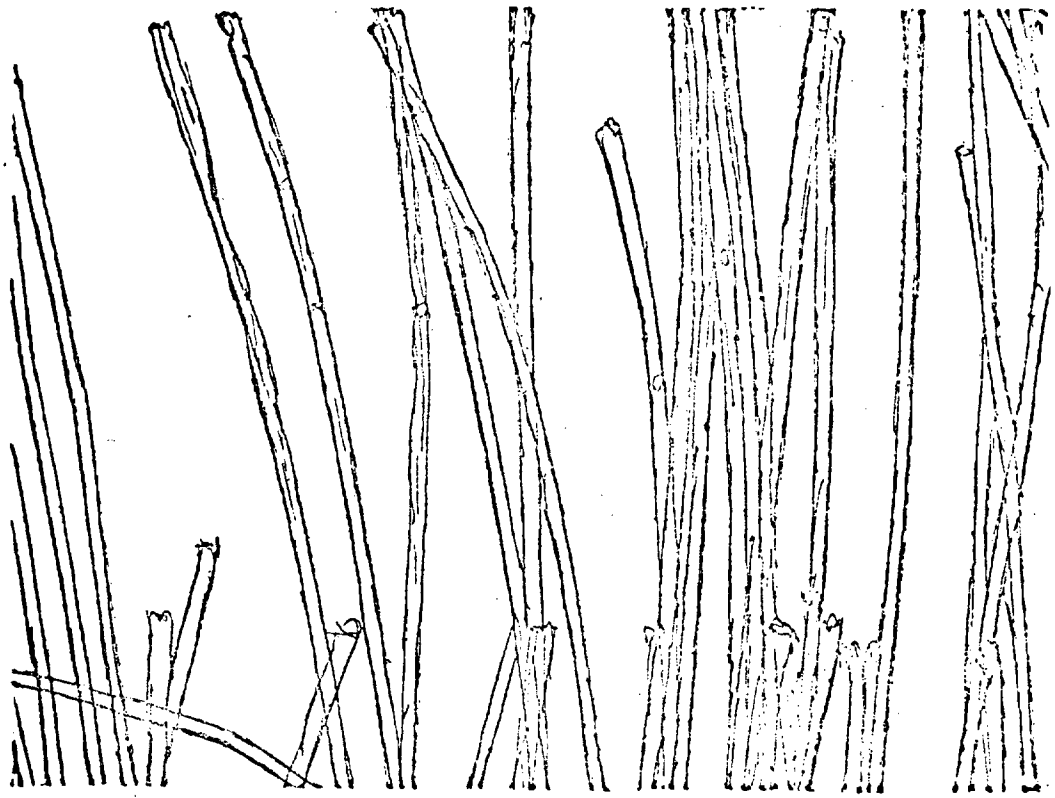


Figure 1A. Photomicrograph of Creslan T-98 before fibrillation

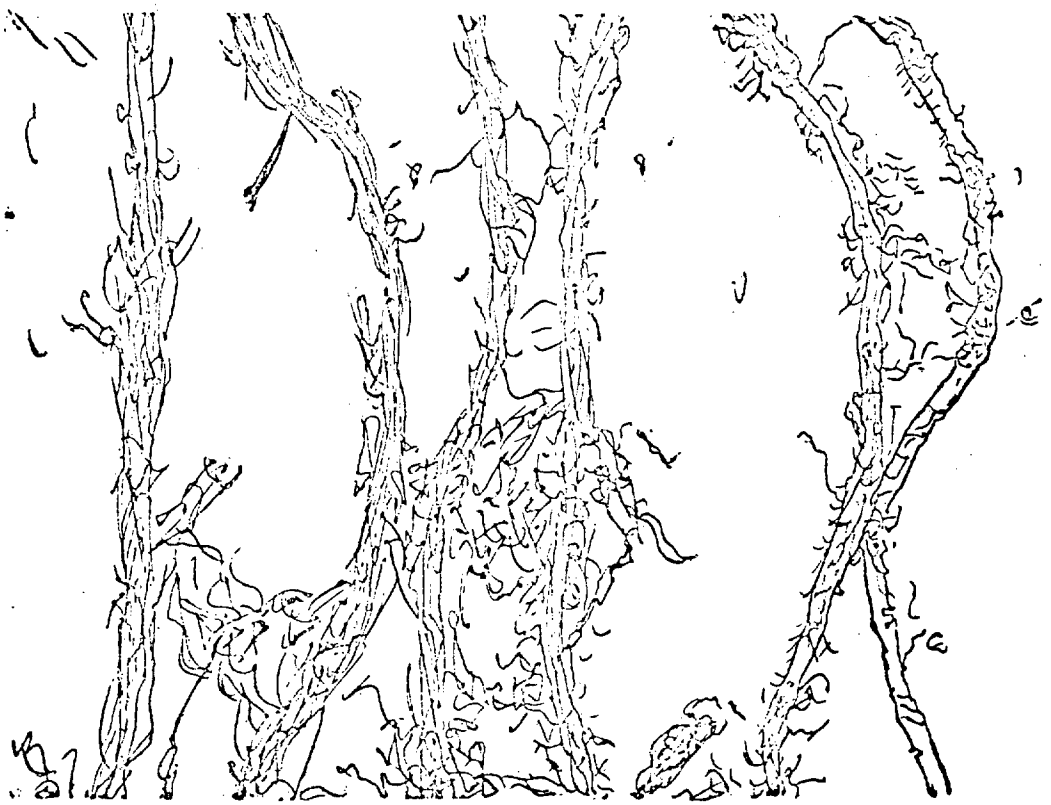


Figure 1B. Photomicrograph of Creslan T-98 after fibrillation

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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